Canopy final interview: Nicole Rycroft

Friday · about 13–14 minutes, then let Nicole steer · clicks in orange brackets

Thank you for making the time, Nicole. I know how full your calendar is.

In my job as a forester, I have a clear guiding light: the forest management plan. Every decision I make, from where we harvest to which values we protect, traces back to it. When I looked at Canopy, I saw the same kind of guiding light: 60 million tonnes of Next Gen by 2033. So I wanted every dataset and every tool I built to trace back to a decision that moves that number.

In my last interview, Keri asked me a question I haven't stopped thinking about: why aren't companies switching to Next Gen faster? I think the honest answer is that it's different for every product. A folding carton for cookies, a corrugated shipping box and a viscose shirt each have different specs, different volumes, different mills and different forests behind them.

So here's my recommendation. Canopy already has the solutions: EcoPaper, the Next Gen providers, Hot Button. But a solution only works when it fits a specific product. It has to match the grade, the volume a mill can actually supply, and the price. And it should go first where the product draws on the most at-risk forest. We can't make that match unless we understand the current demand on the ground: the specific product being made, how much of it, at which mill, and from which forest.

That's what I built the prototype to do. I come at this from the forest end. In the boreal, I follow black spruce from the plan, through harvest, to the mill, and a big share of every tree ends up as pulp, either as pulpwood or as chips from the sawmill. Much of that becomes the packaging Canopy is trying to change. Let me show you.

[Share the browser window: Global overview, Packaging selected.]

This is the prototype. At the top is Canopy's own target: 8.35 million tonnes of Next Gen in 2024, 60 by 2033. That's about 25 percent growth every year. **[Click the "Calculated" badge beside 24.5%.]** Every number in the tool carries its source and a status: reported, estimated or unknown. Click one and you see the formula, the report and the page it came from. If Canopy is going to put a number in front of a brand or an investor, it has to hold up. That's the standard I'd set for everything we build.

[Press Esc. Scroll to "What the 277.9 Mt is made of".] Globally, packaging paper and board is about 278 million tonnes, measured by FAO. Everything under that is an estimate, and Next Gen's share of packaging isn't published at all. So the global picture tells us how big the problem is, but not where to act. To act, we have to go down to companies and products.

[Sidebar: Companies, then Metsä Board.] I started with Metsä Board, because they publish more than most. They make about 1.4 million tonnes of paperboard a year: about half a percent of global packaging. They name nine mills and eight wood-sourcing countries, and 92 percent of their wood is FSC- or PEFC-certified company-wide. That tells us about the company. But Canopy doesn't ask a company to switch "everything". It asks about specific products. So I went one level further.

[Click the product card MetsäBoard Pro FBB Bright.] This is a folding boxboard, the white board you'd see on a cookie box or a cosmetics carton. It's made at the Äänekoski mill in Finland, which has about 260,000 tonnes of capacity: roughly 12.6 percent of the company. The fibre comes from its Environmental Product Declaration. Each tonne is 28 percent bleached chemical pulp and 47 percent BCTMP, all fresh wood fibre, with no recycled content.

[Scroll to "Where the fibre comes from".] And the origin comes from Metsä's EUDR declaration: spruce, pine, birch and aspen from Finland, Sweden, Estonia and Latvia. So we can follow this one product from forest, to pulp mill, to board mill, to the box on the shelf.

[Point at the three "Proxy:" lines.] I'm honest about the gaps. Three of the eight fields are proxies: mill capacity stands in for product volume, the certified share is company-wide, and origin is by country, not by forest. The tool says so, so nobody mistakes an estimate for a fact.

[Scroll down to "Alternative Canopy solutions".] This is where it matters. Once we know the product, we can ask: what could replace it? The tool pulls in candidates: recycled board, agricultural-residue materials, and a bagasse supplier from EcoPaper. But it labels them leads, not matches.

[Sidebar: Canopy solutions, then the EcoPaper tab, with MetsäBoard Pro FBB Bright selected.] Here, EcoPaper listings are ranked against this specific product: does the grade fit cartons and boxes, is there supply near Finland, what evidence is there. And look at what's missing: capacity, minimum order, recycled or Next Gen content, certification.

That's the heart of my point. This product is about 260,000 tonnes a year from one mill. If an alternative can't show it can supply that grade at that volume, it isn't a real alternative for this brand yet. And if no alternative fits, we know where the gap is: what Next Gen Solutions needs to build, or what a provider needs to prove.

[Sidebar: Supply risk.] And the forest decides where we start. This view takes where each producer sources from and checks it against the EU country risk benchmark and Hot Button. Metsä's countries are low-risk under the EU benchmark, and it's highly certified. Sateri, in fashion, comes up high-risk. Country tiers are only a first filter. The next layer I'd add is ForestMapper, so we can see whether a product touches Ancient and Endangered Forests. Products from the most at-risk forests are the ones to move first.

[Sidebar: Disclosure coverage, then click the Mondi row.] Most producers aren't as transparent as Metsä. This view shows how much of each producer's sourcing I could find in public sources. It measures research coverage, not company performance. Metsä has full coverage. Mondi discloses well at company level, but for its SmartKraft Brown product I could only fill one of eight fields. Smurfit Westrock, one of the biggest producers in the world, I haven't researched yet. So the tool writes the ask: publish an EPD and name the mill for the products your brand partners buy, and publish origin by product. From the 30th of December, the EU Deforestation Regulation means large and medium operators have to collect that origin data anyway. Canopy's ask is simply: publish what you already have to collect.

[Sidebar: Global overview, then click Fashion.] Fashion is where I learned the most, because it's so much harder. Canopy's Hot Button covers the whole viscose and lyocell market: about 8.4 million tonnes, and only about one percent comes from recycled feedstock. Hot Button tells us which producers are green, which are at risk, and who sells Next Gen.

[Sidebar: Companies, then Lenzing.] But look at how far tracing gets. Lenzing is a leader. I can see its pulp mills in Austria, the Czech Republic and Brazil, and its pulp output. But the wood origin per product, and its Next Gen share, are unknown.

[Click Sateri (RGE Group).] Sateri holds about a quarter of global capacity, has a high-risk sourcing deduction in Hot Button, and much of its pulp comes from sister companies within RGE. Mills and forest origin aren't public.

In packaging, I can reach the product. In fashion, I hit a wall at the producer. I tried to trace my own shirt back. I even emailed Aditya Birla, one of the more transparent producers, and didn't hear back. That isn't a criticism. It shows how hard this is from the outside. And that's exactly where Canopy's relationships with brands and producers can open doors a researcher can't.

[Sidebar: Next Gen mills, with India selected.] Once you know what a product needs, the next question is: could enough Next Gen supply exist to replace it? Whenever I'm trying to understand a complex issue, I like to start with a back-of-the-envelope calculation. What order of magnitude are we talking about, and could we realistically meet the objective? So I did that for Next Gen mills. You know India far better than I do, so treat this as a test of the logic, not a feasibility study.

[Scroll to the Mill build planner. Point at the three mill-size buttons.] I started with mill size, using three real reference mills: Red Leaf's straw mill in Regina at 200,000 tonnes of pulp a year, Circulose's textile recycling mill in Sweden at 60,000 tonnes, and Infinited Fiber's planned mill in Finland at 30,000 tonnes.

[Point at the feedstock buttons.] Then I looked at the feedstock going in, because India has both kinds. There's about 26 million tonnes of paddy straw a year across Punjab, Haryana and the Delhi region of Uttar Pradesh, much of it still burned in the fields. And there's about 7.8 million tonnes of textile waste a year.

[Click "Haryana paddy straw", then "Red Leaf, straw · 200 kt".] Here's one illustration. Haryana generates about 6.8 million tonnes of paddy straw a year. If 20 percent is collectable, and straw yields about half its weight in pulp, which is Red Leaf's design figure, that supports roughly three mills of 200,000 tonnes each. That's about 0.6 million tonnes of pulp a year: around 40 percent of the 1.5 million tonnes Canopy's India blueprint aims for. At the blueprint's average cost, that's about $0.8 billion of investment, and it avoids about 2.4 million tonnes of CO2 a year at Canopy's four tonnes per tonne of pulp.

[Point at the "If the collectable share is…" row.] But this row is what matters. At 5 percent collectable, it's zero mills. At 50 percent, it's eight. The whole answer hangs on one number nobody has good data for: how

much straw is really collectable once fodder, power plants and logistics take their share. That's the same lesson as packaging: the headline looks big, but the decision depends on the specific detail on the ground.

[Sidebar: Demand and supply.] And the demand side is just as thin. If brands wanted 10 percent of viscose and lyocell to be Next Gen, about 0.7 million tonnes is still missing after today's output and new capacity: roughly twelve mills the size of Circulose's. But no brand publishes its Next Gen demand in tonnes. That's what product-level tracing gives us: which products, how many tonnes, and from which forests, so the case for a mill rests on real demand.

[Stop sharing the screen.]

So, to bring it together. Canopy has the solutions and the relationships. What's missing is the product-level picture: what's being made, how much, at which mill, from which forest. With that, we can say which products to move first, whether an EcoPaper or Next Gen alternative really fits, and how many tonnes brands would need to switch. That last number is what an investor needs before a mill gets built, in India or anywhere else.

In my first year, I'd do four things. First, document the priority datasets: owner, source and update cycle for each, with one shared standard for how we mark evidence. Second, with Pack4Good, trace the products brand partners buy from ten producers, all to the same depth, timed to EUDR. Third, with CanopyStyle, take Hot Button producers down to product lines, starting with the highest-risk sourcing. And fourth, add up what we've traced into tonnes by product type, and pair it with feedstock checks like the one I just showed, so Next Gen Solutions can show investors both real demand and realistic supply. Data and Research would own the evidence; each team would own the decision it supports.

Why me? I know the forest end of this chain first-hand, from the plan to the stump to the mill. And I've shown I can build the tools that connect it to the brands at the other end. I'd love to do that for Canopy.

That's the heart of it. I'm happy to go deeper anywhere you'd like.

If you're running long: skip the Disclosure coverage paragraph (the one that starts with Mondi) and the Lenzing paragraph. That saves about 1.5 minutes. **Then stop talking and let Nicole steer.**